

## July Assessment:

**Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks**

Since the job involves simultaneous coding, and video conferencing, the system must handle multiple resource-intensive tasks simultaneously without lag or slowdown.

- Processor (CPU): we will need a processor with at least 8 cores and fast speed.
  - The more the number of cores, the better they help convert code and run virtual machines faster, while built-in video processing in some CPUs keeps video calls smooth without slowing down your work. Make sure the CPU physically fits your motherboard.
- Memory (RAM): we will need at least 32 GB installed in dual-channel (using two matching RAM sticks instead of one). Coding and video calls take up a lot of memory. Using two sticks instead of one doubles the speed at which the CPU reads data, preventing random breakdown.
- Storage (SSD): Choose a fast NVMe SSD, they provide quick read and write speeds making sure large project files open instantly and your system doesn't slow down or freeze when saving heavy files.
- We should also check that our motherboard supports the RAM speed and has slots, for all our parts. The CPU, RAM and storage should all work well together so our computer does not have any problems.
- We need a Power Supply that gives us more than enough power we need. This is the Power Supply Unit or PSU.

**Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.**

Impact of Background Applications (e.g., Antivirus)

- **Resource Allocation:**

Antivirus is constantly watching the system — scanning files as they're opened, checking programs in memory, and inspecting network traffic. All of this needs CPU processing power, RAM (memory), and disk access — the same resources other apps need and use.

- **Hardware Bottlenecks:**

- **Storage I/O Latency** — When antivirus scans a file in real-time (say, the moment you open or save it), it temporarily "locks up" the disk's read/write queue. So when we try to open, save, or copy a file, it feels sluggish because the antivirus is in line ahead of you, checking things first.

- **RAM Saturation** — If an antivirus is hogging a lot of memory, the computer runs low on free RAM. When that happens, Windows starts shifting the active apps out of fast RAM and into the storage drive (like your HDD/SSD) as "virtual memory." Storage is way slower than RAM, so everything feels laggy — this is called "saturation."
- **CPU Contention** — The CPU can only do so much at once. If it's busy running an antivirus scan, it has less spare capacity for video call or code editor. This shows up as: the CPU usage spikes (can be seen in Task Manager), and the actual app briefly freezes or stutters while the antivirus task gets handled.
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**Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.**

1. Plug the peripheral into different ports (USB 2.0 vs. USB 3.0/Type-C) and also at different operating systems/workstations to confirm whether the issue is hardware- or software-bound
2. Driver Verification: Open Device Manager. Look for warning indicators (yellow exclamation marks).
3. Clean Driver Reinstallation: Uninstall existing drivers, remove legacy software packages, download the latest vendor-signed drivers, and install them.
4. Firmware Update: Update the peripheral's onboard firmware if available from the manufacturer.
5. Sometimes, a connected device (like a printer, mouse, or flash drive) stops working properly because the operating system (Windows/Linux) or background software is interfering with it. Example:
  - a. Power management - To save power sometimes the OS powers off the USB, this may cause the connected device to malfunction.
  - b. Any recent patch update on the OS can also cause malfunction of connected devices.

**Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.**

Uninterruptible Power Supply (UPS): Provides immediate battery backup power during blackouts or brownouts. This allows for automated or manual safe shutdowns, preventing interrupted disk write operations that corrupt file systems.

Surge Protection: prevents physical damage to motherboard ICs, power delivery circuits, and storage controllers due to voltage spikes (transients) caused by power grid switching or lightning strikes.

## Section B: Practical Application

### Task 1: Physically assemble a workstation and document the component installation sequence.

Physical assembly of the ASUS PRIME H410M-CS workstation, step by step, in the order the components were installed.

#### System Overview

Motherboard: ASUS PRIME H410M-CS | CPU: Intel Core i5-10400 (LGA1200) | RAM: Crucial 16GB DDR4-3200 UDIMM | GPU: ASUS GeForce GT 710 (2GB GDDR5) | PSU: Antec VS500L | Storage: SATA HDD/SSD.

#### Preparation Before Assembly

Before installation began, the workspace was cleared and organized, an anti-static wrist strap was worn and clipped to a grounded point, and all tools (magnetic Phillips screwdriver, thermal paste, isopropyl alcohol, lint-free cloth, cable ties, and a parts tray for screws) were laid out within reach. The PSU was kept switched off and unplugged until final power-on,

#### STEP 1 Install the CPU onto the Motherboard

With the motherboard out of the case and the LGA1200 socket's locking arm lifted, the CPU's notch markers were aligned with the socket and lowered straight down without forcing it. The retention bracket was then closed and locked back under the screw to clamp the CPU in place.



#### STEP 2 Install the CPU Fan

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A fresh, even layer of thermal paste was applied to the CPU's heat-spreader lid, replacing the old, cracked paste found during teardown. The fan's four push-pins were aligned with the motherboard's mounting holes, lowered onto the CPU, and each pin was pressed down firmly until it clicked and locked.



### **STEP 3 Install the Motherboard into the Case**

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With the CPU and cooler already attached, the motherboard assembly was lowered onto the case standoffs, the rear I/O ports were aligned with the I/O shield, and the board was secured with its mounting screws.

### **STEP 4 Install the RAM**

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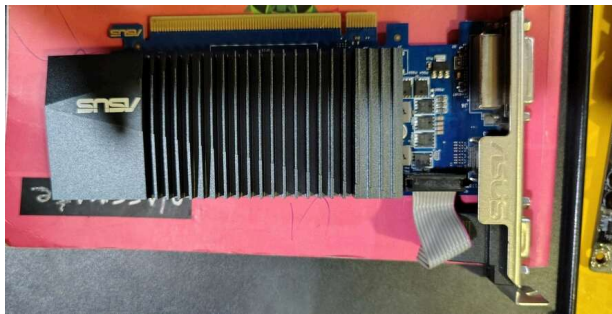
The retention clips on the DIMM slot were opened, the notch on the RAM module was aligned with the slot key, and the module was pressed straight down evenly on both ends until the clips snapped closed.



## **STEP 5 Install the Graphics Card (GPU)**

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The matching case bracket cover was removed, the GPU's PCIe edge connector was aligned with the x16 slot, and the card was pressed down firmly until the retention clip clicked. the bracket was then secured to the case with a screw.



## **STEP 6 Install the Storage Drive (HDD/SSD)**

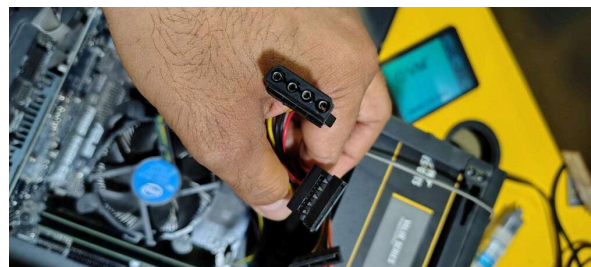
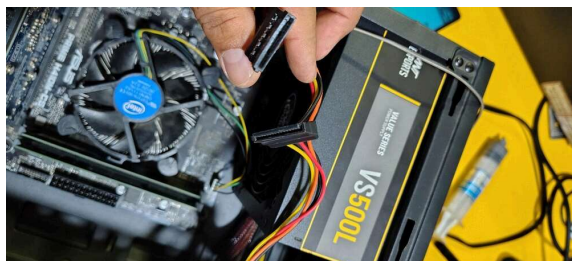
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The drive was slid into its bay or tool-less mount and secured with screws or the case's retention clip, leaving its SATA ports accessible for cabling in the next step.

## **STEP 7 Reconnect SATA Power, SATA Data, and Molex Cables**

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The SATA power cable from the PSU was plugged into the drive, the SATA data cable was connected between the drive and the motherboard, and any spare Molex power connectors were reconnected.

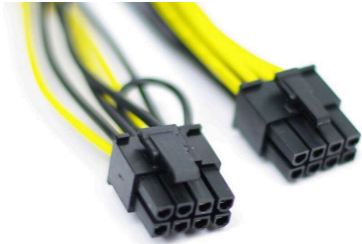




### **STEP 8 Reconnect the CPU 8-Pin (EPS12V) Power Connector**

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The 8-pin EPS12V connector from the PSU was plugged into the motherboard's CPU power socket near the top of the board.



### **STEP 9 Reconnect the Main 24-Pin ATX Power Connector**

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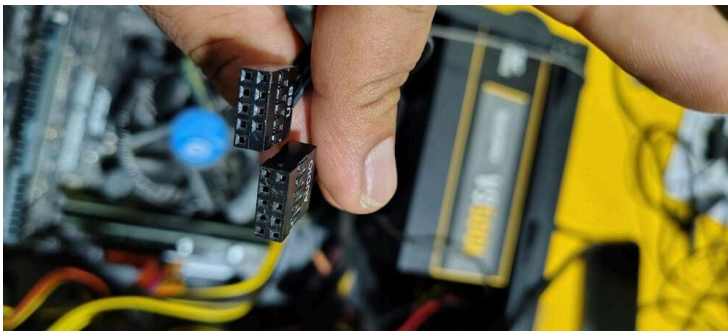
The 24-pin connector's keyed edge was aligned and pressed firmly into the motherboard's main power socket until the retention clip clicked shut.



### **STEP 10 Reconnect the Front-Panel USB and Audio Headers**

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The small "USB" and "AUDIO" connectors, along with the power, reset, and HDD-light leads, were plugged back onto their matching pin headers on the motherboard, checking pin orientation carefully.



## STEP 11 Replace the Side Panel

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Once every component and cable connection was double-checked, the side panel was slid back into place and re-secured with its screws or clips.

## STEP 12 Reconnect Power and Boot

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The power cable was plugged back into the PSU and the wall outlet, the PSU rocker switch was switched back on, and the case's power button was pressed to boot the system.

## Task 2: Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings

**Boot to BIOS/UEFI:** Turn on system and press Del or F2 continuously.

**Verify Memory:** Check the memory summary page to see detected RAM speeds, capacities, and voltages.

- open the computer's BIOS/UEFI screen, go to the main hardware summary page. Look at what the computer actually detects for your RAM sticks.
- **check for:**
  - **Capacity:** check if complete RAM capacity is showing . e.g., If you installed two 16 GB sticks, does it show 32 GB total
  - **Speed (Frequency)**
  - **Voltage:** check if the voltage matches the type of RAM
- If it shows a lower total amount of RAM or strange speeds, one of your RAM sticks might not be seated correctly or might have conflicting settings.

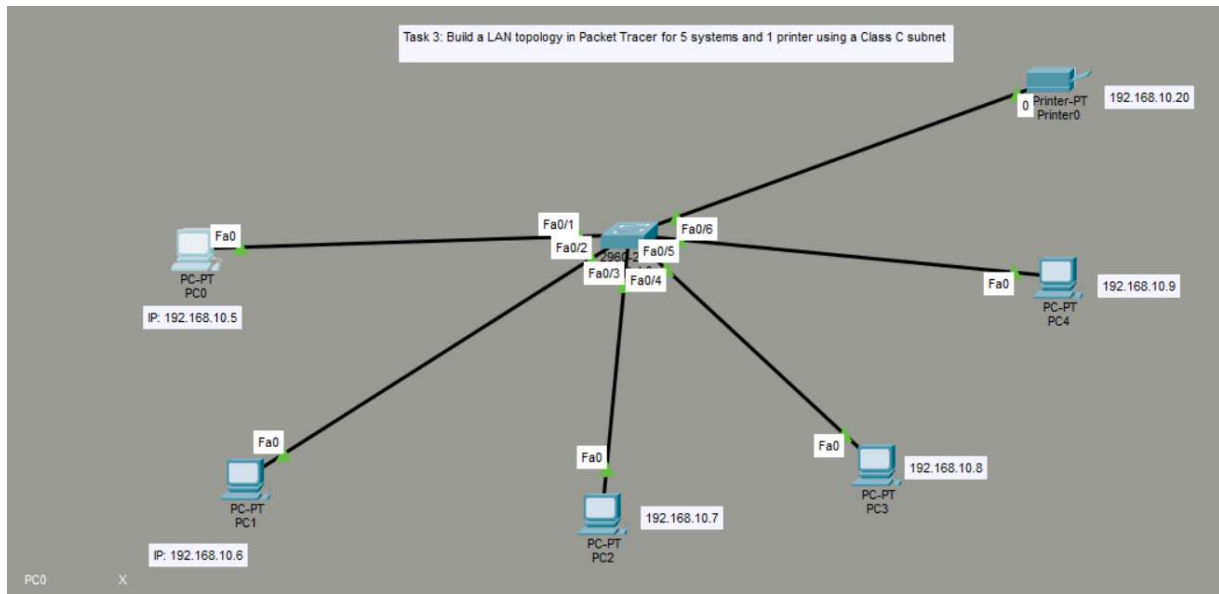
XMP -extreme memory profile and DOCP- Direct over clocking profile are memory overclocking features built into the computer's motherboard.

If 2 RAMS are installed and both are manufactured for different speeds with XMP/DOCP enabled, it will make both RAMS to run at maximum speed, thus making the lower speed RAM to overclock and crash, Now with XMP/DOCP disabled - both RAM are run at lowest speed of the two.

**Manual Configuration:** Manually adjust the memory speed to match the lowest frequency among installed modules, and match the highest stable voltage/latency timings across modules.

**Save & Exit:** Save changes (F10) and restart to test stability.

## Task 3 : Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.



IP Addressing Scheme (Class C Network: 192.168.10.0/24)

- **Network ID:** 192.168.10.0
- **Subnet Mask:** 255.255.255.0
- **Broadcast Address:** 192.168.10.255



```
PCO
Physical Config Desktop Programming Attributes
Command Prompt

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>PING 192.168.10.7

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>PING 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>PING 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time=10ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.9:
```

```
PCO
Physical Config Desktop Programming Attributes
Command Prompt

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>PING 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>PING 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time=10ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>PING 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time=13ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

| Device Name | Device Type       | Interface     | IP Address    | Subnet Mask   |
|-------------|-------------------|---------------|---------------|---------------|
| PC0         | Host PC           | FastEthernet0 | 192.168.10.5  | 255.255.255.0 |
| PC1         | Host PC           | FastEthernet0 | 192.168.10.6  | 255.255.255.0 |
| PC2         | Host PC           | FastEthernet0 | 192.168.10.7  | 255.255.255.0 |
| PC3         | Host PC           | FastEthernet0 | 192.168.10.8  | 255.255.255.0 |
| PC4         | Host PC           | FastEthernet0 | 192.168.10.9  | 255.255.255.0 |
| Printer0    | Network Printer   | FastEthernet0 | 192.168.10.20 | 255.255.255.0 |
| Switch0     | Cisco 2960 Switch | Fa0/1 – Fa0/6 |               |               |

Task 4 : Perform a clean OS installation with manual disk partitioning and startup optimization.

## Windows 10 Installation Guide

### Preparation

#### Step 1 — Check System Requirements

Windows 10 requires the following minimum specifications:

Processor 1 GHz or faster, 64-bit

RAM 2 GB minimum (4 GB+ recommended)

Storage 20 GB minimum (64 GB+ recommended)

Firmware UEFI with Secure Boot support

Check if your laptop supports UEFI

1. Press Win + R, type msinfo32, and hit Enter

2. In System Information, look for BIOS Mode — it should say UEFI

|                             |                                                                             |
|-----------------------------|-----------------------------------------------------------------------------|
| Processor                   | Intel(R) Core(TM) i3-1005G1 CPU @ 1.20GHz, 1190 Mhz, 2 Core(s), 4 Logica... |
| BIOS Version/Date           | Insyde F.57, 9/19/2022                                                      |
| SMBIOS Version              | 3.2                                                                         |
| Embedded Controller Version | 40.38                                                                       |
| BIOS Mode                   | UEFI                                                                        |
| BaseBoard Manufacturer      | HP                                                                          |
| BaseBoard Product           | 85F3                                                                        |
| BaseBoard Version           | 40.38                                                                       |
| Platform Role               | Mobile                                                                      |

#### Step 2 — Back Up Your Data

Back up the following before continuing:

- Personal files: documents, photos, videos to an external drive or cloud storage
- Browser bookmarks (export from your browser's settings menu)
- A list of installed software and product keys (use Belarc Advisor to find keys)
- Email signatures, saved passwords, and custom app configurations

### Step 3 — Create a Bootable USB Drive

- Create bootable pendrive using Media Creation Tool from Microsoft
- Or download the Win 10 ISO from the Microsoft website and use Rufus software to write it. Select GPT partition scheme for UEFI.

### Step 4 — Enter BIOS and Configure Settings

Restart your laptop and press the BIOS key during startup. Common keys by manufacturer: F2, F10/Del

In BIOS, configure the following:

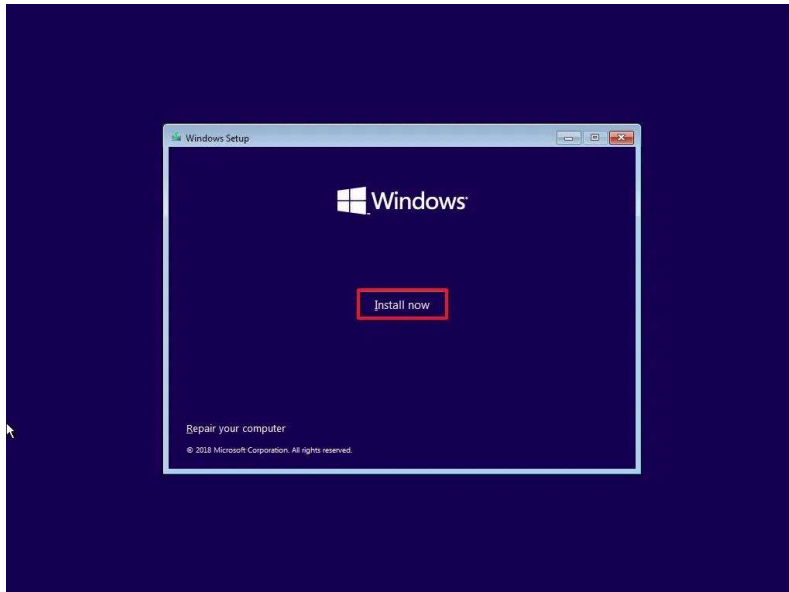
- Enable UEFI boot mode (disable Legacy / CSM if present)
- Enable Secure Boot
- Set the USB drive as the first boot device in the boot order
- Save and exit

### Step 5 — Boot from USB

With the USB inserted and BIOS configured, restart the laptop. The Windows setup screen should appear. If not, press the boot menu key (F12, F9, or Esc depending on your brand) and manually select the USB drive from the list.

### Step 6 — Begin Setup

- Click Install Now on the setup screen



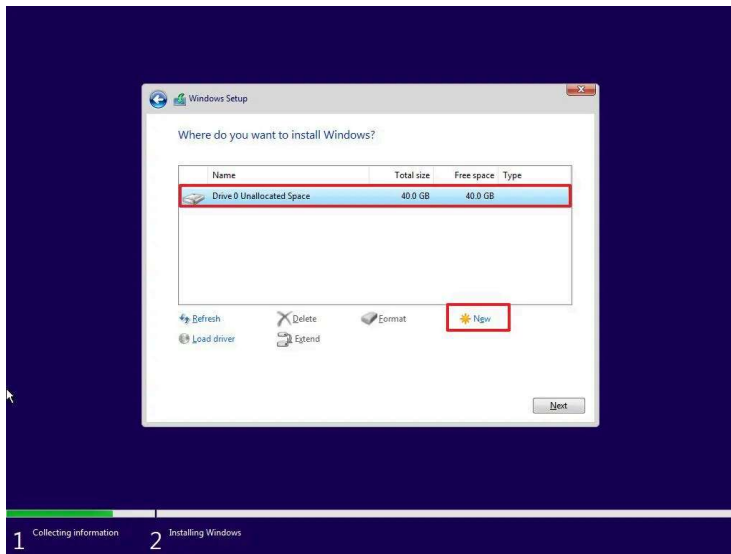
- Enter your product key, or click I don't have a product key to skip (activate later)
- Select your edition: Windows 10 Home for personal use, Pro for advanced features
- Accept the license agreement
- Choose Custom: Install Windows only for a clean install

#### Step 7 — Partition Your Drive

This is the most critical step. You will see a list of drives and partitions.

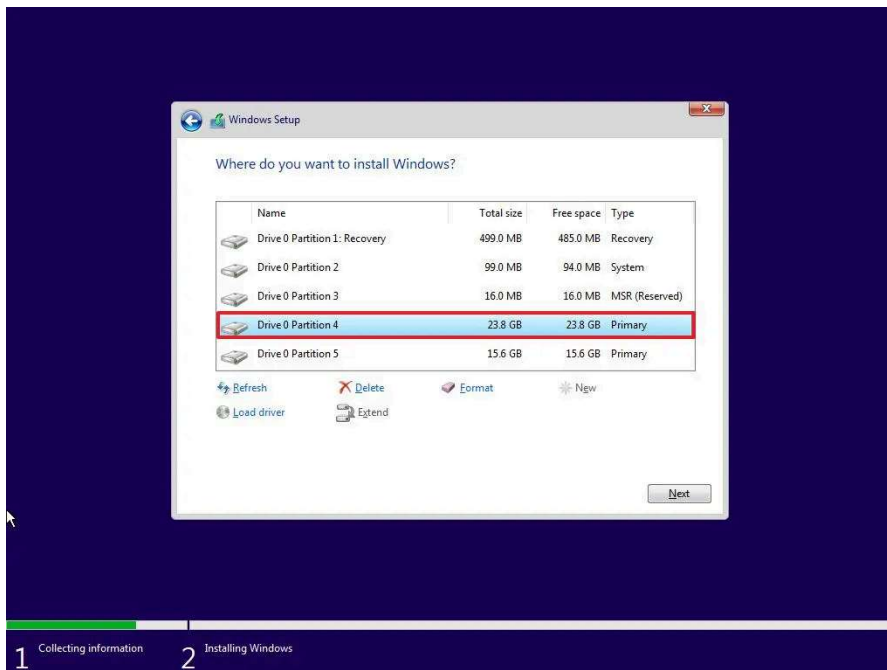
Fresh install (entire drive):

- Delete all existing partitions on the target drive
- Click New — Windows will automatically create the required system partitions
- Create the different partitions required keeping C: drive storage more than minimum required and click Next



Dual boot (keeping an existing OS):

- Do not delete the existing OS partition
- Shrink the existing partition first using Disk Management inside your current OS
- Select the resulting unallocated space and click Next



Step 8 — Wait for Installation to Complete



Windows will copy files, install features, and apply updates. This takes 10–30 minutes depending on hardware speed. The laptop will restart several times. Leave the USB plugged in but do not press any key when prompted to boot from USB.

#### Phase 4: Initial Setup

##### Step 9 — Region, Account & Preferences

- Select your region and keyboard layout
- Connect to Wi-Fi when prompted
- Create a username and password for your account
- Set security questions for local account recovery

##### Step 10 — Privacy During Setup

Windows presents a privacy screen during setup.

##### Step 11 — Run Windows Update

- Go to Settings → Update & Security → Windows Update
- Click Check for updates and install all pending updates
- Restart when prompted — repeat until no updates remain (may take 2–3 cycles)

##### Step 12 — Install Drivers

Windows Update handles most drivers automatically. The drivers latest version can be verified from the manufacturers website.

##### Step 13 — Final Recommended Settings

- Activate Windows: Settings → Update & Security → Activation
- Set a power plan: Control Panel → Power Options → Balanced or High Performance

##### Performance

- Create a restore point: Search Create a restore point → Configure → Create

- Enable Windows Defender: Settings → Windows Security and confirm it is active
- Install essentials: browser, antivirus, productivity and communication apps

## Section C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

Questions:

1. Identify the three hardware components most likely to cause random reboots under load.

**CPU** — under sustained heavy load, an overheating CPU triggers thermal protection shutdowns/restarts, especially if cooling is already weak.

**PSU (Power Supply Unit)** — a PSU that's unable to supply peak voltages, aging, or has degraded capacitors can't sustain the higher current draw of CPU+GPU under full load, causing voltage sag and a sudden reboot.

**GPU** — Heavy workloads make the GPU work hard; poor airflow can cause it to overheat and either throttle hard or trigger a protective shutdown that takes the whole system down with it.

2. Explain why the systems pass POST in the morning but fail in the afternoon.

A Power-On Self-Test (POST) is a built-in diagnostic sequence a computer runs immediately after you press the power button. It verifies core hardware like the CPU, RAM, and storage are functional before handing control over to the operating system.

Neither the CPU nor GPU is pushed anywhere near full load capacity.

In the mornings the average temperature would be low compared to midday temperatures and with poor ventilation and heavy work load the CPU and GPU temperatures would spike causing thermal throttling.

### 3. How would you use a stress-test tool to confirm a thermal or power issue?

Below are screen shot of image taken during Stress test carried out using 'cinebench' and monitored using 'HWmonitor', to explain how stress test will help isolate the problem.

| Sensor                          | Value      | Min       | Max        |
|---------------------------------|------------|-----------|------------|
| HALO                            |            |           |            |
| HP 85F3                         |            |           |            |
| Temperatures                    |            |           |            |
| \TZ.TZ02                        | 10.0 °C    | 10.0 °C   | 10.0 °C    |
| \TZ.TZ00                        | 28.0 °C    | 28.0 °C   | 28.0 °C    |
| Utilization                     |            |           |            |
| Physical Memory Load            | 94.0 %     | 92.0 %    | 94.0 %     |
| Available Physical Me...        | 0.4 GB     | 0.4 GB    | 0.6 GB     |
| Virtual Memory Load             | 93.0 %     | 92.5 %    | 93.2 %     |
| Available Virtual Mem...        | 0.8 GB     | 0.8 GB    | 0.9 GB     |
| Intel Core i3 1005G1            |            |           |            |
| Voltages                        |            |           |            |
| VID (Max)                       | 0.912 V    | 0.585 V   | 1.025 V    |
| Temperatures                    |            |           |            |
| Cores (Max)                     | 92.0 °C    | 63.0 °C   | 100.0 °C   |
| Package                         | 99.0 °C    | 60.0 °C   | 100.0 °C   |
| Powers                          |            |           |            |
| Package                         | 15.23 W    | 4.37 W    | 18.00 W    |
| IA Cores                        | 11.72 W    | 1.81 W    | 14.47 W    |
| GT                              | 0.01 W     | 0.00 W    | 0.45 W     |
| Power Max (PL1)                 | 17.00 W    | 17.00 W   | 17.00 W    |
| Short Power Max (PL2)           | 35.00 W    | 35.00 W   | 35.00 W    |
| Power Limit 3 (PL3)             | 48.00 W    | 48.00 W   | 48.00 W    |
| Max Peak Power (PL4)            | 101.00 W   | 101.00 W  | 101.00 W   |
| Currents                        |            |           |            |
| IccMax                          | 54.00 A    | 54.00 A   | 54.00 A    |
| Clocks                          |            |           |            |
| Cores (Max)                     | 2893.6 MHz | 196.5 MHz | 3412.0 MHz |
| CPU BCLK                        | 99.8 MHz   | 49.1 MHz  | 100.6 MHz  |
| LLC/Ring                        | 3192.9 MHz | 196.5 MHz | 3218.1 MHz |
| Memory                          | 1065.7 MHz | 524.1 MHz | 1065.7 MHz |
| Utilization                     |            |           |            |
| Processor                       | 212.0 %    | 16.7 %    | 219.7 %    |
| TOSHIBA MQ04ABF100 ( 70IDPESUT) |            |           |            |
| Temperatures                    |            |           |            |
| Assembly                        | 40.0 °C    | 40.0 °C   | 40.0 °C    |
| Counters                        |            |           |            |
| Power-On-Hours                  | 6172 hrs   | 6172 hrs  | 6172 hrs   |
| Power Cycle Count               | 3959       | 3959      | 3959       |
| Utilization                     |            |           |            |
| Used Space (d:)                 | 17.2 %     | 17.2 %    | 17.2 %     |
| Available Capacity (d:)         | 205.2 GB   | 205.2 GB  | 205.2 GB   |
| Used Space (e:)                 | 10.5 %     | 10.5 %    | 10.5 %     |
| Available Capacity (e:)         | 227.2 GB   | 227.2 GB  | 227.2 GB   |
| Used Space (f:)                 | 31.6 %     | 31.6 %    | 31.6 %     |
| Available Capacity (f:)         | 293.8 GB   | 293.8 GB  | 293.8 GB   |

#### Confirming Thermal Issues :

- **Diagnostic Procedure:** Run an isolated CPU load (e.g., Cinebench, Prime95) or GPU stress test (e.g., FurMark) while tracking temperatures, fan behavior, and clock speeds using HWmonitor

- **Evidence in Image:** As shown in the screenshot, thermal failure is confirmed when package and core temperatures hit their critical limit—in this case, reaching **100.0deg.C Max**.

#### Clock speed:

- **Cores (Max):** ranged from **196.5 MHz** (min) up to **3412.0 MHz** (max)
- A modern CPU doesn't normally idle-clock as low as 196.5 MHz during an active load test — dropping to ~200 MHz mid-run is the CPU's thermal protection kicking in and slashing frequency because it hit its temperature ceiling. This is the classic example of **thermal throttling**, not a power-delivery fault.

| POWER X NET1000 (DWP2309256K3377)            |           |           |             |
|----------------------------------------------|-----------|-----------|-------------|
| <b>Temperatures</b>                          |           |           |             |
| Assembly                                     | 54.0 °C   | 54.0 °C   | 54.0 °C     |
| <b>Counters</b>                              |           |           |             |
| Power-On-Hours                               | 2849 hrs  | 2849 hrs  | 2849 hrs    |
| Power Cycle Count                            | 2447      | 2447      | 2447        |
| Health Status                                | 97 %      | 97 %      | 97 %        |
| <b>Utilization</b>                           |           |           |             |
| Used Space (c:)                              | 57.1 %    | 57.1 %    | 58.5 %      |
| Available Capacity (c:)                      | 101.9 GB  | 98.4 GB   | 101.9 GB    |
| Activity                                     | 2.2 %     | 0.0 %     | 60.2 %      |
| <b>Speed</b>                                 |           |           |             |
| Read Rate                                    | 0.19 MB/s | 0.00 MB/s | 71.95 MB/s  |
| Write Rate                                   | 0.24 MB/s | 0.00 MB/s | 113.41 MB/s |
| <b>Intel(R) UHD Graphics</b>                 |           |           |             |
| <b>Utilization</b>                           |           |           |             |
| GPU                                          | 2.5 %     | 0.4 %     | 27.5 %      |
| Video Engine                                 | 0.0 %     | 0.0 %     | 0.0 %       |
| 3D                                           | 2.5 %     | 0.4 %     | 27.5 %      |
| Copy                                         | 0.0 %     | 0.0 %     | 0.0 %       |
| Compute                                      | 0.0 %     | 0.0 %     | 0.0 %       |
| Dedicated Memory                             | 0.0 MB    | 0.0 MB    | 0.0 MB      |
| Shared Memory                                | 221.4 MB  | 127.0 MB  | 249.7 MB    |
| <b>PABAS0241231</b>                          |           |           |             |
| <b>Voltages</b>                              |           |           |             |
| Current Voltage                              | 12.206 V  | 12.068 V  | 12.206 V    |
| <b>Powers</b>                                |           |           |             |
| Charge Rate                                  | 30.05 W   | 28.07 W   | 30.21 W     |
| <b>Capacities</b>                            |           |           |             |
| Designed Capacity                            | 36000 mWh | 36000 mWh | 36000 mWh   |
| Full Charge Capacity                         | 36000 mWh | 36000 mWh | 36000 mWh   |
| Current Capacity                             | 23310 mWh | 20490 mWh | 23310 mWh   |
| <b>Levels</b>                                |           |           |             |
| Wear Level                                   | 0.0 %     | 0.0 %     | 0.0 %       |
| Charge Level                                 | 64.8 %    | 56.9 %    | 64.8 %      |
| <b>VirtualBox Host-Only Ethernet Adapter</b> |           |           |             |
| <b>Speed</b>                                 |           |           |             |
| Download Speed                               | 0.00 Mbps | 0.00 Mbps | 0.00 Mbps   |

**Current Voltage: 12.068–12.206V** — stayed tight and stable throughout the sample, with no sag or spike. A power-delivery fault would show voltage dropping noticeably (or fluctuating erratically) right around the time of the thermal event; instead it barely moved.

#### 4. Propose three specific steps to remediate the frequent system restarts.

**Improve airflow immediately** — change the location of the units out of the enclosed desk cabinets or add active ventilation: cabinet exhaust fan, vented panels, or simply leave doors open and reapply thermal paste if it's old/dried.

**Clean and re-seat cooling components** — dust out CPU heatsinks, GPU fans, and case fans (dust buildup is a very common silent cause of thermal throttling), and verify all case/CPU fans are actually spinning at expected RPM under load.

**Validate and if needed upgrade the PSU** — run the CPU and GPU stress test together and watch the 12V rail in HWiNFO. If the voltage dips noticeably or the PSU is just too weak for what the CPU and GPU draw together at full load, swap it for a higher-wattage, good-quality unit that has some power to spare above that peak draw.

#### 5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

A PSU upgrade probably isn't needed if the voltage stays steady during the stress test and the reboots line up with the CPU/GPU temperature hitting the shutdown point instead. In that case the PSU is working fine, and the real fix is better airflow and cooling, not more wattage.

Since these units sit in poorly ventilated cabinets and only crash after a long period of heavy use — not right at the start — that points to heat building up over time, not a power dip. So overheating is the more likely cause here, and upgrading the PSU should only be considered after the stress test confirms it's actually needed, not as the first fix.